Leakage-Aware Deep Learning for Financial Event-State Forecasting

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DL Course Project

Historical research artifact only: no trading advice, no live forecast, no positive benchmark claim.

Talk Roadmap

Question

- Can aligned company-news event states improve multi-horizon excess-return forecasts?
- Can deep models help without leaking future information?

Answer

- The strongest active evidence is an optimal model XGBoost event-market baseline.
- The optimal mixed model (frozen-base horizon mixer) improved validation but not optimal model XGBoost lockbox.
- The main contribution is a rigorous, reproducible evaluation system.

Why This Is Hard

- Returns have low signal-to-noise and strong nonstationarity.
- News can matter, but timestamps, no-news days, and provider truncation create hidden biases.
- Model selection can overfit validation folds even when deep models look plausible.

Design principle

Train/validation is for search. Lockbox is for one final post-selection diagnostic.

Metric principle

ROC-AUC is not applicable: this is continuous excess-return regression.

Protocol: Leakage-Aware Artifact Pipeline

Stage	Artifact	Key guardrail	Output
Data	Top-50 + OHLCV + news	no raw overwrite	row universe
Features	market + event states	no-news rows preserved	sidecars
Splits	expanding folds + lockbox	purge/embargo	split manifest
Training	fold-local preprocessing	no lockbox tuning	saved preds
Evaluation	RMSE/MAE + rank IC	bootstrap + claim gate	reports

- Horizons: 1d, 5d, 10d, 20d, 60d.
- Lockbox ends 2026-02-13; all result roots keep manifests and prediction packets.

State of the Art Context

Financial ML baselines

- Efficient-market caution and expected-return multiple testing.
- Gradient boosting is still a very strong tabular baseline.
- Forecast metrics: RMSE, MAE, OOS R^2, rank IC.

Deep learning candidates

- FinBERT/BERT for semantic financial text features.
- Optimal model TSMixer and Transformer-family sequence models.
- PatchTST, iTransformer, TFT, and Chronos motivate richer temporal architectures.

Implemented Model Stack

Text/Event

- Alpha Vantage aligned news.
- No-news as explicit state.
- FinBERT sidecars for later fusion.

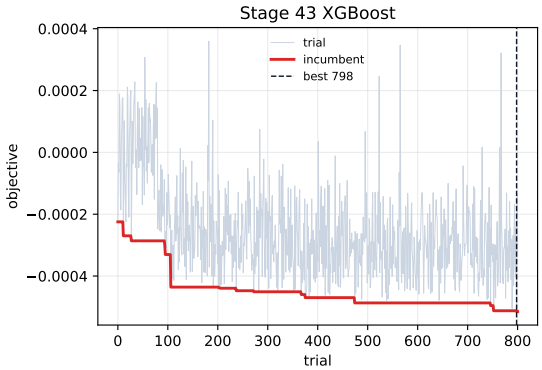
Market

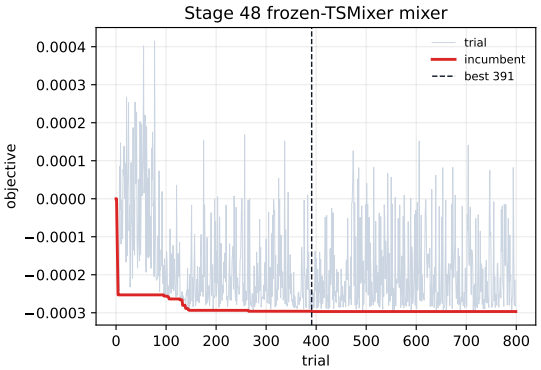
- Lag/rolling features.
- Linear and fixed XGBoost baselines.
- Optimal model XGBoost: 800-trial tuned baseline.

Sequence/Mixer

- TSMixer diagnostics.
- Readout correction audit.
- Optimal mixed model (frozen-base horizon mixer): horizon mixer only.

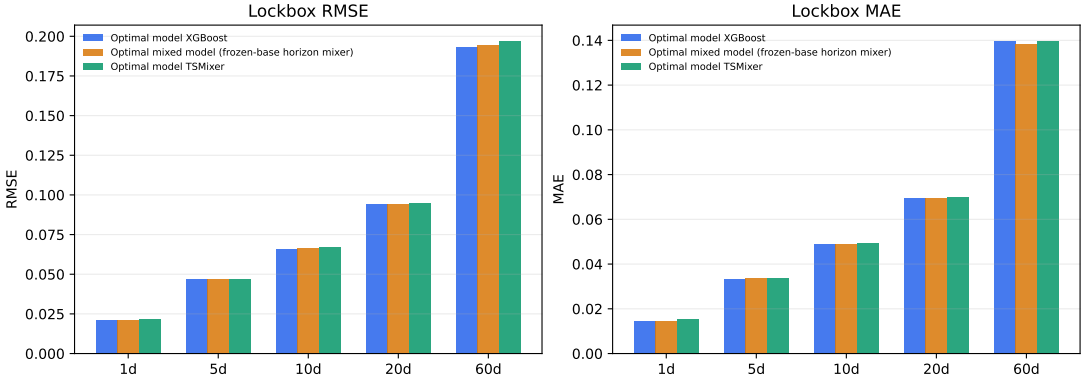
Tuning Trace: Search Was Auditable





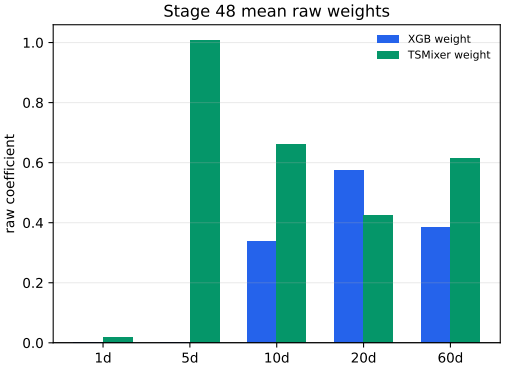
- Optimal model XGBoost: 800 completed trials, best trial 798.
- Optimal mixed model (frozen-base horizon mixer): 800 completed meta/mixer trials, best trial 391.
- Both searches used zero lockbox rows for tuning.

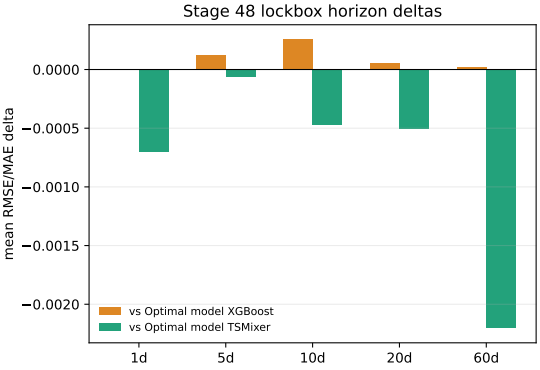
Primary Metrics: Lockbox RMSE and MAE



- Regression metrics are primary because labels are continuous forward excess returns.
- The optimal mixed model does not dominate optimal model XGBoost on lockbox.

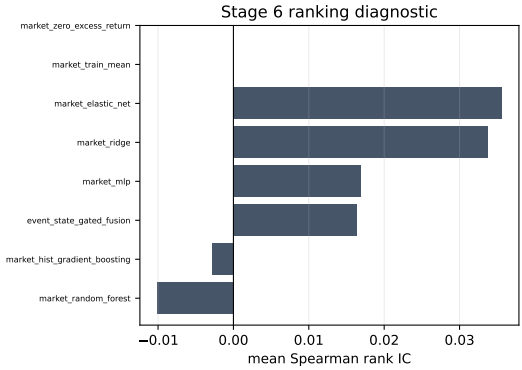
Deep Learning Lesson: Validation Gains Are Not Enough

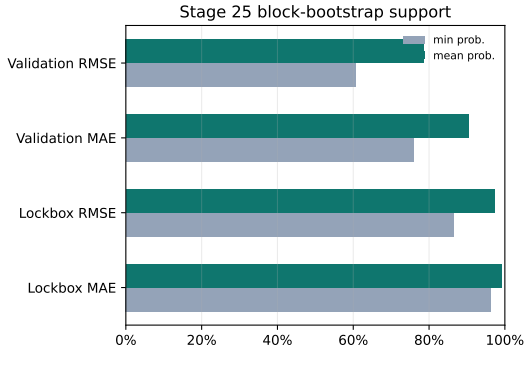




- The optimal mixed model improved validation weighted error vs optimal model XGBoost by -2.97×10^{-4}.
- It worsened post-selection lockbox weighted error vs optimal model XGBoost by $+9.20 \times 10^{-5}$.
- Conclusion: stronger architectures require stronger selection controls, not looser claims.

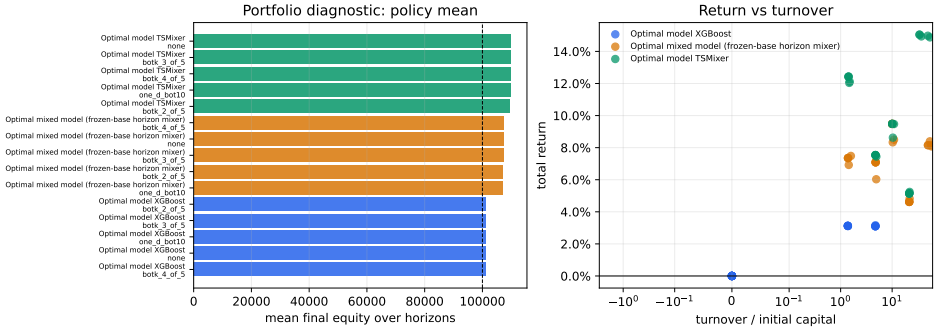
Secondary Evidence: Ranking and Uncertainty





- Rank IC and hit-rate diagnostics are secondary: useful for interpretation, not primary selection.
- Moving-block bootstrap checks temporal dependence and uncertainty.

Portfolio Simulation: Diagnostic Only



- Initial capital: \$100k; long-only, fractional shares, no costs, no slippage, no taxes.
- Best historical row: \$115,045 final equity, but it uses frozen optimal model TSMixer diagnostic predictions.
- This is not a buy/sell system and not benchmark claim evidence.

Main Numerical Takeaways

Evidence	Result	Interpretation
Optimal model XGBoost	lockbox delta -5.34×10^{-4}	active strongest evidence
Optimal model XGBoost	62.2% both-metric improvement	non-claim improvement evidence
Optimal mixed model	validation delta -2.97×10^{-4}	tuned signal looked useful
Optimal mixed model	lockbox delta $+9.20 \times 10^{-5}$	did not beat optimal model XGBoost lockb
ROC-AUC	not reported	target is continuous regression
Portfolio	diagnostic only	no trading/signal claim

Discussion: What I Learned

Positive lessons

- Reproducibility can be engineered through manifests, saved predictions, and tests.
- Tree baselines are hard to beat in small financial tabular settings.
- Deep sequence models need architecture audits and strong baselines.

Cautions

- Validation improvement did not guarantee lockbox improvement.
- Portfolio-looking plots can mislead without transaction realism and causal validation.
- Data compilation is a bottleneck: API limits (25 calls/day) slow dataset refresh; webscraping is advised to expand coverage.

Final Conclusion

Scientific conclusion

The optimal model XGBoost event-market pipeline is the current strongest artifact-backed evidence, but event-state value and deep-sequence superiority are not proven.

Deep learning conclusion

For financial forecasting, the most important DL competence is not just adding BERT, optimal model TSMixer, or Transformers; it is controlling leakage, validating architecture semantics, and resisting overfit narratives.

Performance and simulation conclusion

Performance gains are modest and fragile across horizons, and portfolio simulations remain diagnostic-only; the strongest evidence is the optimal model XGBoost lockbox deltas, not the historical portfolio curves.

THANK YOU